Highlights

Hyper-Heuristic Enhancement of Holistically-Nested Edge Detection

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- ILS-HED enhances a frozen HED backbone via a searchable pool of 98 classical detector configurations.
- Achieves annotation-free domain adaptation through Iterated Local Search and a lightweight fusion layer.
- Gains tend to grow with domain divergence (FID), an exploratory trend ($n = 5$).
- Failure-case analysis with a gradient-variance screening rule guides real-world deployment.

Hyper-Heuristic Enhancement of Holistically-Nested Edge Detection

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ABSTRACT

Deep edge detectors, such as Holistically Nested Edge Detection (HED), tend to degrade when deployed on out-of-domain images. However, restoring performance by retraining the backbone requires extensive labelled datasets and significant graphics processing unit (GPU) computational time. This study proposes **ILS-HED**, a selection-perturbative hyper-heuristic that enhances a frozen HED backbone by automatically selecting and fusing complementary classical edge detectors without backbone parameter updates or target domain annotation. An Iterated Local Search (ILS) algorithm explored a pool of 98 discrete classical detector configurations of the Canny, Sobel, Laplacian, and Gabor families. The response of the best-selected configuration was fused with the frozen HED multiscale side outputs using a lightweight and learnable layer. On the most domain-shifted benchmark, DeepCrack, ILS-HED yields an F1 improvement of +0.0270 over the frozen HED baseline ($p < 0.001$; Cohen's $d = 1.21$ over per-image scores), and it performs comparably to fine-tuned RCF and BDCN while leaving all backbone weights unchanged. We observed consistently positive improvements on all five datasets. The largest gains appeared on the two datasets furthest from the BSDS500 training distribution, as measured by the Fréchet Inception Distance, although with five datasets, this trend was exploratory and not strictly monotonic. This suggests that classical operators provide the greatest complement in regimes where deep natural-image priors are least effective. A failure case analysis revealed that semantically well-defined regions and compressed images are the main conditions under which ILS-HED delivers only marginal improvements, thereby guiding its future application. ILS-HED is an effective deployment-efficient domain adaptation method because it recovers much of the performance gap between frozen and fine-tuned models without retraining, making it useful in medical, infrastructure, and industrial inspection applications where annotation resources are scarce.

1. Introduction

Edge detection forms the basis of numerous computer vision applications, such as object recognition, scene understanding, medical image analysis, and industrial inspection, by providing a geometric framework that supports higher-level reasoning (Canny, 1986; Ziou and Tabbone, 1998). Deep learning has improved the boundary recall of natural images. Holistically Nested Edge Detection (HED) (Xie and Tu, 2017) introduced deep multi-scale supervision over a VGGNet backbone, and its successors, such as RCF (Liu et al., 2019a), BDCN (He et al., 2022), DexiNed (Soria et al., 2020), and CATS (Huan et al., 2022), have continued to improve the benchmark scores on the BSDS500 dataset (Arbeláez et al., 2011). A comprehensive survey of this progression was conducted by Sun et al. (2022).

Despite these advances, three limitations constrain their practical applications. First, the internal fusion weights of the HED were fixed after training. They cannot adapt to new image statistics, whether they are high-texture natural scenes, low-contrast retinal fundus images, or homogeneous industrial surfaces. Second, out-of-domain deployment causes accuracy degradation, and recovering performance via full fine-tuning requires backpropagation through millions of parameters and large annotated target domain data sets. Parameter-efficient fine-tuning (PEFT) methods, such as LoRA adapters (Hu et al., 2022), visual prompt tuning (Jia et al., 2022), and bottleneck adapters (Houlsby et al., 2019), reduce the parameter burden but still require annotations and modify the effective computation of the model backbone. Third, classical operators such as Canny, Sobel, Laplacian, and Gabor remain effective in high-contrast regions, fine-texture contexts, and structured domains. However, no framework systematically searches the configuration space to determine the optimal complement for a given target domain.

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We propose **ILS-HED** as a deployment-efficient solution to these three limitations: a selection-perturbative hyper-heuristic that enhances a frozen HED backbone without modifying its parameters or requiring target domain annotations. The key insight is that classical operators provide maximal complementary value precisely where frozen deep priors are weakest, that is, in domains distant from the BSDS500 natural image distribution. An Iterated Local Search (ILS) algorithm explores 98 classical detector configurations, and a lightweight learnable fusion layer integrates the selected response with HED's frozen multi-scale side outputs.

Compared with full fine-tuning and other PEFT methods, ILS-HED sacrifices some in-domain accuracy in exchange for a fundamentally different strength: fast, annotation-free adaptation that keeps the backbone completely untouched by the task. This makes it suitable for practitioners who must deploy an existing HED model across diverse imaging modalities without the resources required for complete retraining. The framework is deliberately scoped to gradient-defined boundaries in high-quality imagery, the conditions under which classical operators are informative, and its limitations in semantic-boundary or compressed-imagery contexts are explicitly characterised in our failure case analysis.

This study makes four contributions to extant literature.

1. **A hyper-heuristic formulation of classical–deep edge fusion.** We cast the selection of classical edge detectors and their hyperparameters as a combinatorial optimisation problem solved via Iterated Local Search, situating this within the PEFT and adaptive preprocessing literature and identifying the gap that gradient-free external adaptation addresses.
2. **A frozen-HED enhancement strategy for domain adaptation.** ILS-HED adapts HED to new domains using only an ILS search and a small fusion layer, enabling deployment without annotated target data. Extending it to other backbones is left for future work.
3. **Cross-domain empirical validation with rigorous statistics.** We evaluated five heterogeneous datasets and compared them against frozen and fine-tuned HED, RCF, BDCN, and a LoRA-based PEFT baseline, ground the domain-divergence hypothesis using FID measurements, and reported Wilcoxon signed-rank tests (Wilcoxon, 1945) with effect sizes and 95% confidence intervals throughout.
4. **Failure-case analysis with deployment guidance.** We characterised the conditions under which ILS-HED offers limited benefits, thereby providing practitioners with actionable decision criteria.

The remainder of this paper is organised as follows. Section 2 reviews classical and deep-edge detection. Section 3 defines the hyper-heuristics. Section 4 surveys the relevant literature. Section 5 formalises the three target limitations. Section 6 describes the proposed ILS-HED methodology. Section 7 presents the quantitative, qualitative, and ablation results, including a failure case study. Section 8 summarises the findings and future directions.

2. Background

2.1. Edge Detection

An edge marks the boundary between two distinct image regions, identified by sudden changes in intensity, colour, or texture (Amer and Sameer, 2015). For a continuous image function $I(x, y)$, the edge points correspond to local maxima of the gradient magnitude $\nabla I = \left[\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y}\right]^T$ or equivalently, the zero crossings of the Laplacian $\nabla^2 I$ (Marr and Hildreth, 1980; Canny, 1986). Edge-based segmentation exploits these discontinuities to identify region boundaries, and accurate edge maps directly determine the quality of downstream segmentation, feature matching, and quantitative measurements (Ziou and Tabbone, 1998; Haralick and Shapiro, 1985; Ma and Manjunath, 2000).

2.2. Traditional Edge Detection

2.2.1. First-Order Derivative Operators

First-order methods approximate the image gradient to identify intensity transitions. The Roberts operator (Roberts, 1965) uses 2×2 diagonal kernels as follows:

$$G_x = \begin{bmatrix} +1 & 0 \\ 0 & -1 \end{bmatrix}, \quad G_y = \begin{bmatrix} 0 & +1 \\ -1 & 0 \end{bmatrix} \tag{1}$$

Its compact footprint renders it susceptible to noise. The Prewitt operator (Prewitt, 1970) extends to 3×3 kernels to suppress noise modestly. The Sobel operator (Sobel, 1968) adds anisotropic weighting as follows

$$G_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix} \tag{2}$$

approximating the derivative of a Gaussian function and improving the noise immunity. All first-order operators produce wide edge responses that require thinning and exhibit directional bias (Gonzalez and Woods, 2018).

2.2.2. Second-Order Derivative Operators

Second-order methods detect edges as zero-crossings of the Laplacian $\nabla^2 I = \frac{\partial^2 I}{\partial x^2} + \frac{\partial^2 I}{\partial y^2}$. The discrete 3×3 kernel is

$$L = \begin{bmatrix} 0 & +1 & 0 \\ +1 & -4 & +1 \\ 0 & +1 & 0 \end{bmatrix} \tag{3}$$

When applied directly, the Laplacian severely amplifies noise. The Laplacian of Gaussian (LoG) (Marr and Hildreth, 1980) resolves this via Gaussian pre-smoothing, as follows:

$$\mathrm{LoG}(x, y) = -\frac{1}{\pi\sigma^4}\left(1 - \frac{x^2 + y^2}{2\sigma^2}\right) e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{4}$$

where σ controls the noise-detail trade-off. Marr and Hildreth (Marr and Hildreth, 1980) showed that LoG zero crossings correspond to physically meaningful intensity variations.

2.2.3. The Canny Edge Detector

The Canny detector (Canny, 1986) implements four stages: (1) Gaussian smoothing with standard deviation σ; (2) gradient magnitude and orientation computation; (3) non-maximum suppression to thin edges to a single-pixel width; and (4) hysteresis thresholding, retaining pixels above a high threshold T_h as confirmed edges and those between T_l and T_h only when connected to the confirmed edges. Canny remains the classical benchmark in domains where interpretability and deterministic behaviour are required.

2.2.4. Gabor Filters

Gabor filters model orientation-selective responses and are particularly effective for detecting directional structures, such as blood vessels and cracks (Jain and Farrokhnia, 1991). A Gabor filter at orientation θ, wavelength λ, and bandwidth b responds maximally to sinusoidal intensity oscillations aligned with θ. Their multi-orientation responses make them the preferred classical complement for thin directional structures.

2.3. Deep Learning Approaches

CNN-based edge detectors learn feature hierarchies directly from annotated boundaries. HED (Xie and Tu, 2017) demonstrated that deep supervision at multiple network scales substantially improves boundary recall in natural images. Subsequent work includes RCF (Liu et al., 2019a), which exploits all convolutional layers within each VGG stage; BDCN (He et al., 2022), which propagates semantic information bidirectionally across scales; EDTER (Pu et al., 2022), which uses transformer self-attention for long-range dependency modelling; DexiNed (Soria et al., 2020), a lightweight architecture with dense extreme inception blocks for thin-edge detail; and CATS (Huan et al., 2022), which introduces texture-suppression aggregation. A comprehensive survey of deep edge detection methods was provided by Sun et al. (2022). More recent advances include DiffusionEdge (Ye et al., 2024), which applies denoising diffusion probabilistic models to boundary prediction, and EdgeSAM (Zhou et al., 2025), a lightweight on-device distillation of the Segment Anything Model that enables efficient boundary inference without a full SAM deployment. This highlights that the frontier has shifted toward foundation model adaptation since the introduction of HED.

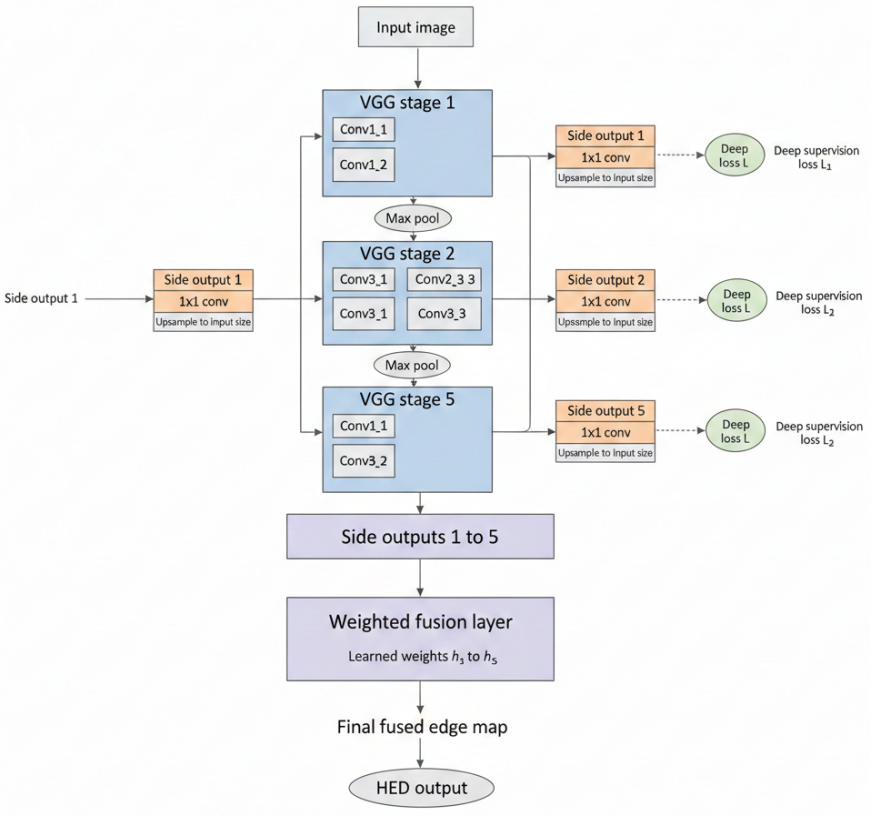


Figure 1: Holistically-Nested Edge Detection architecture. Side-output layers attached at multiple depths within a truncated VGGNet produce progressively coarser edge predictions, combined with a learned weighted fusion layer. Fusion weights are fixed after training; ILS-HED augments these frozen outputs externally.

2.3.1. Holistically-Nested Edge Detection

HED reframes edge detection as an image-to-image prediction task in which the model transforms a full input image into a full output image of the same spatial dimensions, learning a direct mapping from pixels to pixels. Side-output layers attached at multiple depths within a truncated VGGNet (Simonyan and Zisserman, 2014) produce edge predictions at progressively coarser resolutions, combined with a learned weighted-fusion layer (Figure 1). Early layers capture fine-textural details, whereas deeper layers encode semantic context. A class-balanced cross-entropy loss addresses the severe class imbalance between edge and non-edge pixels, and deep supervision regularises training by injecting gradients into intermediate layers.

2.3.2. RCF and BDCN

RCF (Liu et al., 2019a) extends HED by exploiting all convolutional layers within each VGG stage to produce richer multi-scale representations. BDCN (He et al., 2022) further refines fusion by propagating semantic information bidirectionally through a cascade of scale-specific supervisions. Both methods achieve higher benchmark scores than HED but require full end-to-end retraining on the target domains, incurring the same adaptation burden described in Section 5. These serve as strong baselines for our evaluation (Section 7).

3. Hyper-heuristics

A **heuristic** is a problem-solving strategy that guides the search toward good solutions when exact methods are computationally intractable (Tan et al., 2004). A **hyper-heuristic** is a high-level automated search methodology that solves complex optimisation problems by selecting or generating lower-level heuristics rather than directly searching the solution space (Pillay, 2020; Burke et al., 2013). The defining characteristic is that the search operates in the *space of heuristics*, promoting generality across problem instances.

3.1. Classification

Following the taxonomy of Burke et al. (2013), hyper-heuristics are classified along two axes: whether they *select* from existing heuristics or *generate* new ones, and whether the selected heuristics are *constructive* or *perturbative* heuristics. This yields four classes of hyper-heuristics.

1. **Selection-constructive**: select low-level construction heuristics at each step when building a solution from scratch.
2. **Selection-perturbative**: select low-level perturbative heuristics to improve existing solutions, comprising heuristic selection and move acceptance components.
3. **Generation-constructive**: create new constructive heuristics via problem attributes and genetic programming.
4. **Generation-perturbative**: generate new perturbative heuristics by combining existing ones.

ILS-HED belongs to the **selection-perturbative** class: it selects from a predefined pool of classical edge detector configurations to improve the HED edge map, and its acceptance criterion determines whether to retain each newly evaluated configuration.

4. Related Work

4.1. Hyper-heuristics in Optimisation and Vision

Selection-perturbative hyper-heuristics are well established in combinatorial domains, such as scheduling, bin packing, and vehicle routing, where no single heuristic dominates all instances (Cowling et al., 2001, 2000). In these settings, adaptive selection consistently outperformed the static commitment to a single method, motivating the same principle for image processing. In computer vision, the image content, noise conditions, and illumination vary markedly across instances; operators such as Canny perform well in high-contrast regions, whereas deep models excel when contextual reasoning is required (Ozcan et al., 2008). The application of adaptive heuristic selection to frozen deep edge-detection backbones is a specific gap addressed in this study.

4.2. Edge Map Fusion Strategies

Existing ensemble strategies range from static weight averaging (Ma and Manjunath, 2000) to deep fusion layers with fixed and post-training parameters (Liu et al., 2019a). HED (Xie and Tu, 2017) learns scalar fusion coefficients for each side output during training; these weights are frozen at the test time, preventing instance-specific adaptation. BDCN (He et al., 2022) propagates semantic information bidirectionally across scales but requires end-to-end retraining. EDTER (Pu et al., 2022) uses transformer self-attention to modulate multiscale features at a high cost. DexiNed (Soria et al., 2020) achieved a competitive thin-edge performance with a lightweight architecture. CATS (Huan et al., 2022) addresses texture confusion by suppressing the aggregation. None of these methods provide a lightweight, annotation-free policy that selects among heterogeneous classical and deep learning detectors without backbone modification.

4.3. Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) methods provide practical alternatives to full-backbone retraining. LoRA (Hu et al., 2022) injects trainable low-rank matrices into frozen weight layers, achieving competitive performance with a small number of parameters. Visual prompt tuning (VPT) (Jia et al., 2022) prepends learnable tokens to input sequences. Bottleneck adapters (Houlsby et al., 2019) insert small modules within the frozen transformer blocks. These methods reduce the training cost substantially but still require target domain annotations and, in the case of LoRA and adapters, modify the effective computation of the backbone model. ILS-HED differs fundamentally in that its adaptation is entirely external to the backbone and operates on a library of classical preprocessing operators. This makes ILS-HED complementary to PEFT: a LoRA-adapted backbone can serve as the frozen deep component of future hybrid systems.

4.4. Learned and Differentiable Augmentation

A related thread concerns differentiable augmentation pipelines and neural preprocessing modules. AutoAugment (Cubuk et al., 2019) searches for augmentation policies that maximise validation accuracy via reinforcement learning. RandAugment (Cubuk et al., 2020) simplifies this process to a two-parameter random search. Faster AutoAugment (Hataya et al., 2020) and DADA (Li et al., 2020) introduced a differentiable augmentation search. Neural architecture search (NAS) (Zoph et al., 2018) explores structural preprocessing. ILS-HED occupies a complementary

niche: rather than learning augmentation policies for training-time data augmentation, it searches for the classical edge operator that best complements a frozen backbone at deployment time. The discrete, combinatorial structure of the heuristic space makes the metaheuristic search more natural than gradient-based optimisation; the pool elements are non-differentiable operators (Canny hysteresis, Laplacian zero-crossings) with no continuous relaxation.

4.5. Test-Time Adaptation and Source-Free Domain Adaptation

Test-time adaptation (TTA) and source-free domain adaptation (SFDA) address the same core problem as ILS-HED: adapting a pre-trained model to a new target domain without access to source data or target-domain annotations. TTT (Sun et al., 2020) adapts model parameters at inference time using an auxiliary self-supervised task, whereas TENT (Wang et al., 2021) minimises prediction entropy over batch statistics during deployment. More recent SFDA methods (Liang et al., 2020; Yang et al., 2021) transfer domain-invariant features using pseudo-labels or information-maximization objectives without requiring target annotations.

ILS-HED occupies a delineated and specialised position within the broader conceptual and methodological landscape. Existing TTA and SFDA approaches adapt *internal* model parameters, such as batch normalisation statistics, feature extractor weights, or classifier heads; therefore, they require gradient flow through the backbone at the time of deployment. ILS-HED is completely external to the backbone. No backbone gradients are computed, no parameters are modified, and adaptation operates over a discrete library of classical preprocessing operators rather than the model-weight space. This makes ILS-HED complementary rather than competitive. A TENT-adapted backbone can serve as the frozen deep component of ILS-HED, potentially combining internal entropy minimisation with external classical complementation. The gradient-free, annotation-free nature of ILS-HED is its defining constraint and primary practical advantage in settings where even lightweight inference-time backpropagation is prohibited by hardware or latency requirements.

5. Problem Statement

Contemporary deep-edge detectors suffer from three interrelated limitations that constrain their practical deployment in real-world applications.

Problem 1: Fixed Fusion and Neglect of Classical Complements. HED, RCF, and BDCN fuse multi-scale side outputs using weights learned during training (Xie and Tu, 2017; Liu et al., 2019a; He et al., 2022). Once the optimisation is concluded, these fusion coefficients are static. These pipelines rely exclusively on learned hierarchical features and provide no mechanism for injecting complementary classical cues from operators that excel in high-contrast, thin-structure, and directional-texture contexts, where deep natural image priors oversmooth or miss fine details.

Problem 2: Domain Sensitivity and Adaptation Burden. HED backbones trained on BSDS500 degrade when deployed on out-of-domain data, such as retinal fundus images, infrastructure crack photography, and industrial stone segmentation. Full fine-tuning is computationally and annotation intensive. PEFT methods reduce the parameter count but still require annotations and modification of the backbone computation. Both methods prevent rapid adaptation without additional annotations.

Problem 3: Absence of Automated Classical-to-Deep Orchestration. Although classical operators have been manually combined with deep outputs in ad hoc ensembles, no existing framework systematically searches the configuration space of classical detectors to discover the optimal complements for a given domain. With 98 discrete parameter combinations across four detector families, an exhaustive offline search is feasible but does not exploit the local parameter structure, and applying a single fixed classical setting across all images is suboptimal.

6. Proposed Approach: ILS-HED

ILS-HED enhances a frozen HED backbone by treating classical edge detector selection and hyperparameter configuration as a combinatorial optimisation problem solved using Iterated Local Search. Because the backbone parameters remain fixed, ILS-HED functions as a deployment-efficiency method, recovering much of the performance difference between frozen and fine-tuned models without requiring additional annotations or backpropagation through the deep layers.

6.1. Architecture Overview

The framework comprises three interacting components (Figure 2).

1. **Frozen HED backbone.** A pre-trained HED network with VGG-style encoding and deep-supervised side outputs produces multi-scale edge probability maps $\{S_1, \ldots, S_5\}$. All the backbone parameters were fixed. Only the side-output fusion layer was exposed to adaptation.
2. **Classical heuristic pool.** A discrete library $\mathcal{H}$ of 98 classical edge-detection configurations. Each element defines a complete low-level heuristic that maps the input image to an edge response map.
3. **Hyper-heuristic controller.** An ILS algorithm that adaptively selects and perturbs configurations, evaluates each against a fitness function, and trains a lightweight fusion module by combining the best classical response with HED side outputs.

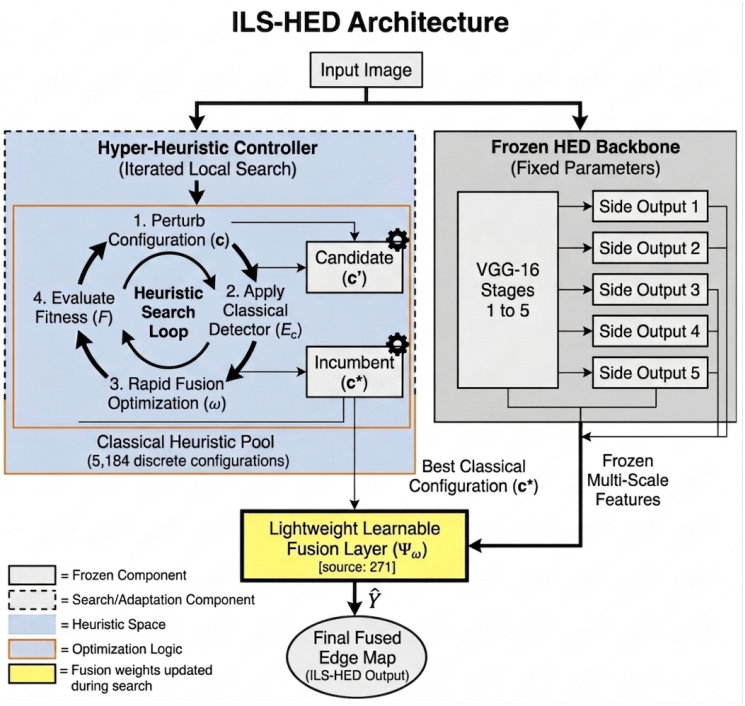


Figure 2: ILS-HED architecture. The hyper-heuristic controller searches across 98 classical detector configurations. The selected classical response is fused with the frozen HED side outputs via learnable scalar weights. No backbone parameters are modified.

6.2. Low-Level Heuristic Pool

The four detector families in $\mathcal{H}$ were chosen for their failure characteristics, which complement those of deep natural-image priors, and their hyperparameter ranges were set to cover the typical operating regimes for each family reported in the edge-detection literature (Canny, 1986; Sobel, 1968; Marr and Hildreth, 1980; Jain and Farrokhnia, 1991), as follows:

- **Canny:** $\sigma \in \{0.5, 1.0, 1.5, 2.0\}$, $T_l \in \{10, 25, 50, 75\}$, $T_h \in \{100, 150, 200\}$. Canny provides precise thin-edge localisation and is particularly effective for high-contrast structures.
- **Sobel:** Kernel size $k \in \{3, 5\}$, threshold $\tau \in \{50, 100, 150, 200\}$. Sobel captures macro-scale gradient boundaries with a directional specificity.
- **Laplacian:** Kernel size $k \in \{3, 5\}$, scale $\sigma \in \{0.5, 1.0, 1.5\}$, threshold $\tau \in \{50, 100, 150\}$. LoG zero-crossings provide second-order boundary localisation, which is effective for dark-line features on uniform backgrounds.
- **Gabor:** Orientations $\theta \in \{4, 8\}$, frequency $\lambda \in \{3, 5, 7\}$, bandwidth $b \in \{0.5, 1.0\}$, threshold $\tau \in \{50, 100\}$. Multi-orientation Gabor responses detect thin, directional structures, such as vessels and cracks.

The Cartesian product of each family's parameter settings yielded the following configuration counts:

- **Canny:** $|\sigma| \times |T_l| \times |T_h| = 4 \times 4 \times 3 = 48$ configurations.
- **Sobel:** $|k| \times |\tau| = 2 \times 4 = 8$ configurations.
- **Laplacian:** $|k| \times |\sigma| \times |\tau| = 2 \times 3 \times 3 = 18$ configurations.
- **Gabor:** $|\theta| \times |\lambda| \times |b| \times |\tau| = 2 \times 3 \times 2 \times 2 = 24$ configurations.

The aggregate pool comprises $48 + 8 + 18 + 24 = 98$ unique configurations across the four families. The pool is intentionally restricted to interpretable, computationally lightweight operators to ensure fast ILS evaluation (0.01–0.04s per configuration on a standard CPU). The effect of the pool composition was assessed in an ablation study (Section 7.9).

6.3. Iterated Local Search as Hyper-Heuristic

The configuration pool $|\mathcal{H}| = 98$ was discrete and non-differentiable. Exhaustive evaluation is computationally feasible offline ($98 \times 0.03\,\text{s} \approx 3\,\text{s}$). However, three considerations motivate the use of ILS over a naive exhaustive sweep.

First, **per-image deployment speed**: ILS with $T_{\max} = 50$ evaluates fewer than half the pool on any given image while concentrating on evaluations in the most promising subspace via targeted perturbations. This is important for the configuration predictor (Phase 2), which is trained on the ILS-selected configurations. A predictor trained on structured ILS trajectories generalises better than one trained on uniform exhaustive samples, as confirmed by the ablation results in Section 7.9 (a random search over 50 samples yields a lower F1 score than ILS at the same budget).

Second, **search quality**: ILS exploits the local structure of each family's parameter space through parameter-level perturbation (probability 0.7), converging to the exhaustive optimum at approximately half the evaluation budget (Section 7.9). The ablation results in Table 8 demonstrate that greedy hill climbing, ILS without adaptive perturbation, and full ILS provide incremental gains, confirming that the iterative perturbation-and-acceptance loop adds value beyond simple enumeration.

Third, **Scalability**: The 98-configuration pool was intentionally minimal. Future extensions incorporating additional operator families (structured forests, orientation-energy filters, and tiny task-specific CNNs) will substantially increase the pool size, at which point the scalability advantage of ILS over the exhaustive search will become significant. The framework was designed to accommodate this growth without architectural changes.

ILS is particularly appropriate for this task because it maintains only one incumbent solution, applies focused perturbations to avoid becoming trapped in local optima, and offers a balance between performance and evaluation cost in combinatorial heuristic selection (Lourenço et al., 2003). A comparison of ILS with random search and greedy hill-climbing in this specific space is included in the ablation study (Section 7.9).

6.3.1. Main Algorithm

Algorithm 1 presents the ILS-HED procedure used in this study. The algorithm maintains the current best configuration c^* and iterates through perturbation, local search (fusion optimisation), and acceptance until the computational budget is exhausted.

6.3.2. LocalSearchFusion Sub-Procedure

Algorithm 2 details fusion optimisation, which constitutes the local search step of each ILS iteration.

The fusion output is

$$\hat{Y} = \sigma\left(\sum_{l=1}^{5} w_l S_l + w_{\text{cls}} \cdot E_{c'} + b\right), \tag{5}$$

where $\sigma(\cdot)$ is the sigmoid function and $\omega = \{w_1, \ldots, w_5, w_{\text{cls}}, b\}$ are the seven learnable scalar parameters. The fusion parameters were optimised using Adam (Kingma and Ba, 2015) with HED's class-balanced cross-entropy loss. To avoid circular contamination between the ILS fitness signal and the fusion weight optimisation, each image is partitioned into two *disjoint* tiles: a **fitness tile** $\mathcal{T}_F$ (5% of pixels, used only by the ILS acceptance criterion in Equation 6)

Algorithm 1 ILS-HED Hyper-Heuristic

Require: Image I, frozen HED Φ_{HED}, heuristic pool $\mathcal{H}$, fusion network Ψ_ω, fitness F, max iterations $T_{\max}$, stagnation threshold κ, random seed r

SetSeed(r)
$c_0 \leftarrow \text{RandomConfiguration}(\mathcal{H})$
$E_0 \leftarrow \text{Apply}(c_0, I)$
$c^* \leftarrow c_0$; $s^* \leftarrow \text{LocalSearchFusion}(c_0, I, \Phi_{\text{HED}}, \Psi_\omega)$; stag $\leftarrow 0$
for $t = 1, \ldots, T_{\max}$ **do**
    $c' \leftarrow \text{Perturb}(c^*, \text{stag}, \kappa)$
    if $\sigma^2_\nabla(I) < 0.01$ **and** c' is Laplacian with $|\tau| > 150$ and $\sigma < 0.5$ **then**
        continue ▷ Optional FM3 filter: reject ringing-prone configs on low-gradient images
    end if
    $s' \leftarrow \text{LocalSearchFusion}(c', I, \Phi_{\text{HED}}, \Psi_\omega)$
    if $F(s') > F(s^*)$ **then**
        $c^* \leftarrow c'$; $s^* \leftarrow s'$; stag $\leftarrow 0$
    else
        stag $\leftarrow$ stag $+ 1$
    end if
end for
return s^*, c^*, ω^*

Algorithm 2 LocalSearchFusion

Require: Configuration c', image I, frozen HED Φ_{HED}, fusion parameters ω, learning rate η, epochs E, fitness tile $\mathcal{T}_F$, training tile $\mathcal{T}_W$ (disjoint: $\mathcal{T}_F \cap \mathcal{T}_W = \emptyset$)

$E_{c'} \leftarrow \text{Apply}(c', I)$
$\{S_1, \ldots, S_5\} \leftarrow \Phi_{\text{HED}}(I)$ ▷ Frozen forward pass
Initialise $\omega' \leftarrow \omega$
for epoch $= 1, \ldots, E$ **do**
    $\hat{Y} \leftarrow \sigma\Big(\sum_{l=1}^{5} w_l S_l + w_{\text{cls}} \cdot E_{c'} + b\Big)$
    $\mathcal{L} \leftarrow \text{ClassBalancedCE}(\hat{Y}, Y_{\mathcal{T}_W})$
    $\omega' \leftarrow \omega' - \eta \nabla_{\omega'} \mathcal{L}$
end for
return $\hat{Y}$, ω'

and a **training tile** $\mathcal{T}_W$ (5% of pixels, used only for fusion weight gradient updates). The remaining 90% of the pixels constitute the test region, which is never observed during either the search or weight optimisation. This partition is fixed per image for the duration of all experiments. Because fusion is a linear combination of pre-computed maps, backpropagation never touches the frozen backbone.

6.3.3. Perturbation Operator

Two perturbation strategies are employed, selected by a uniform random variable:

1. **Parameter-level perturbation (probability 0.7):** Randomly alter one hyperparameter of the current detector family while retaining the family (e.g., increment σ or swap threshold values). This exploits the local structure of each family's parameter space.
2. **Family-level perturbation (probability 0.3):** Replace the detector family entirely and sample a random valid parameter vector for the new family. This enables a global exploration of the detector types.

When stagnation exceeds κ iterations without improvement, the perturbation switches unconditionally to the family level to escape the local optima.

6.3.4. Fitness Function

For datasets with ground truth annotations, the fitness of a fused edge map s is the ODS F-score, as follows:

$$F(s) = \frac{2 \cdot \text{Precision}(s, Y_{\text{GT}}) \cdot \text{Recall}(s, Y_{\text{GT}})}{\text{Precision}(s, Y_{\text{GT}}) + \text{Recall}(s, Y_{\text{GT}})}. \tag{6}$$

This was evaluated on the fitness tile $\mathcal{T}_F$ (5% of image pixels, disjoint from the fusion weight training tile $\mathcal{T}_W$) rather than the full image for computational efficiency. The correlation between the tile-level and image-level fitness was validated empirically in Section 7.8.

For annotation-free cross-domain transfer, a composite no-reference fitness combining edge coherence, boundary continuity, and spurious response suppression is used (Ziou and Tabbone, 1998).

6.3.5. Acceptance Criterion

Strict improvement acceptance is employed: c' replaces c^* only if $F(s') > F(s^*)$. This greedy criterion is standard for ILS in combinatorial heuristic selection and avoids the risk of degradation under a probabilistic acceptance schedule with a limited evaluation budget.

6.4. Training and Deployment Protocol

The protocol operates in two phases:

Phase 1 (Hyper-Heuristic Search). For each training image or tile, the ILS was executed for $T_{\max}$ iterations. The best configuration c^* and optimised fusion weights ω^* were recorded. Each run used a fixed seed drawn from a predetermined schedule to ensure reproducibility (seeds 0–4 for five-run experiments). No backpropagation occurred through the backbone layers.

Phase 2 (Configuration Predictor). The recorded (I, c^*) pairs train a lightweight *configuration predictor*—a three-layer convolutional network (Conv-32, Conv-64, GlobalAvgPool, FC-128, FC-5 for family classification; a parallel FC-regressor for continuous hyperparameters). Training uses cross-entropy loss for the family head and mean squared error for the regressor, with Adam (Kingma and Ba, 2015) at a learning rate 10^{-3} for 30 epochs on the training split images. The predictor reduces the deployment-time adaptation to a single forward pass. A comprehensive evaluation of the predictor's performance and its accuracy relative to the full ILS search is presented in Section 7.7.

7. Experiments

This section evaluates ILS-HED across five heterogeneous datasets. Three hypotheses were tested: (H1) classical–deep fusion improves over a frozen HED baseline; (H2) gains increase with domain divergence from natural images, as measured by Fréchet Inception Distance (FID) from BSDS500; and (H3) ILS search discovers transferable selection patterns without backbone retraining.

7.1. Datasets and Evaluation Protocol

FID values (Table 1) were computed between the BSDS500 training images and each target dataset using InceptionV3 features, providing an objective measure of domain divergence that was independent of ILS-HED outputs.

BSDS500 (Arbeláez et al., 2011) is the canonical benchmark for natural-image edge detection, with 500 images and multiple human-annotated boundaries. The ODS F-score is computed by fixing a single contour threshold for the entire test set, and OIS allows per-image adaptation. Average precision (AP) is the area under the precision-recall curve over all thresholds. Precision and recall used a distance tolerance of 0.0075.

DRIVE (Staal et al., 2004) contains 40 colour retinal fundus images from diabetic retinopathy screening. Ground truth consists of manual vascular tree segmentations. Thin, curvilinear vessels with low contrast test whether classical operators recover the fine details suppressed by frozen HED priors.

STARE (Hoover et al., 2000) contains 20 retinal images under varied conditions including pathological cases with lesions. Leave-one-out cross-validation (LOO) was performed following standard practices.

DeepCrack (Liu et al., 2019b) contains 537 training and 200 test images of asphalt and concrete cracks—long, thin, discontinuous targets with highly variable contrast on homogeneous but noisy backgrounds. This represents the most extreme domain departure from the natural images (FID = 89.4).

Stone331 (Huang et al., 2020) contains 331 stone-surface images for materials segmentation. Irregular, texture-dominated boundaries with ill-defined ground truths constitute challenging industrial test cases.

Table 1
Dataset characteristics, evaluation metrics, and FID from BSDS500 training distribution. FID was computed using InceptionV3 features; a higher FID indicates greater domain divergence. LOO: leave-one-out cross-validation. BSDS500 image counts are given as train/val/test.

Dataset	Domain	Images	Resolution	Metric	FID↑
BSDS500 (Arbeláez et al., 2011)	Natural scenes	200/100/200	481×321	ODS F-score	0.0
DRIVE (Staal et al., 2004)	Retinal fundus	20/20	565×584	F1-score	48.3
STARE (Hoover et al., 2000)	Retinal fundus	20 (LOO)	700×605	F1-score	51.7
DeepCrack (Liu et al., 2019b)	Pavement cracks	537/200	varied	F1-score	89.4
Stone331 (Huang et al., 2020)	Stone segmentation	331 (5-fold)	480×320	F1-score	83.1

7.2. Implementation Details

All experiments were conducted using PyTorch 1.12 with CUDA 11.3 on an NVIDIA RTX 3090 (24GB VRAM). The HED backbone is a publicly available VGG-16 model pretrained on BSDS500 and ImageNet, with all backbone parameters frozen. The classical heuristic pool was implemented using OpenCV 4.5 and Scikit-image 0.19.

Reproducibility. The DRIVE, STARE, DeepCrack, Stone331, and small-annotation experiments were repeated five times with seeds $\{0, 1, 2, 3, 4\}$. The BSDS500 experiments and ablation study (Table 8) were repeated three times with seeds $\{0, 1, 2\}$. We used fewer seeds for these two because the seed-to-seed standard deviation on BSDS500 was already at most 0.001 ODS for every method, and the ablation compared configurations against each other under identical seeds; therefore, additional runs would not change its ordering. The number of runs is also mentioned in each table caption. Across runs, both the ILS initialisation (random c_0) and fusion-layer Adam initialisation were re-seeded; therefore, the reported standard deviations captured the combined stochasticity of the search process and fusion optimisation. The Canny and Sobel baselines involve no randomness; therefore, their run-to-run standard deviation is zero for every dataset.

ILS hyperparameters. $T_{\max} = 50$ for BSDS500 and $T_{\max} = 100$ for smaller medical and industrial datasets. Stagnation threshold $\kappa = 10$. Perturbation probabilities: parameter-level $p = 0.7$ and family-level $p = 0.3$. Fusion weight optimisation uses Adam with a learning rate 10^{-3} for 20 epochs on the training tile $\mathcal{T}_W$ (5% of image pixels). The fitness tile $\mathcal{T}_F$ (a separate, disjoint 5% of pixels) is used exclusively by the ILS acceptance criterion. These two partitions are non-overlapping; see Section 6.3 for full details.

LoRA baseline. LoRA adapters (Hu et al., 2022) were originally proposed for attention weight matrices in transformers, but have been adapted to convolutional settings by treating a 1×1 convolutional layer as a low-rank matrix product (He et al., 2023). We injected LoRA matrices (rank $r = 4$, $\alpha = 8$) into the HED fusion layer modelled as 1×1 convolutions, following this convention. The adapter is trained on each target-domain training split with the same Adam optimiser and early stopping as ILS-HED's fusion layer (∼12,000 additional trainable parameters). Backbone weights remain frozen. We acknowledge that this is an approximation; the results should be interpreted as a directional PEFT comparison rather than a definitive LoRA benchmark. We used a single rank ($r = 4$) and did not test sensitivity to r, so the annotation crossover in Table 9 applies to this setting.

Baselines:

1. Canny (default): $\sigma = 1.0$, $T_l = 50$, $T_h = 150$.
2. Sobel (default): kernel $k = 3$, $\tau = 100$.
3. HED (frozen): pre-trained backbone, no adaptation.
4. HED (fine-tuned): full parameter update on target-domain training split.
5. RCF (Liu et al., 2019a): frozen and fine-tuned modes.
6. BDCN (He et al., 2022): frozen and fine-tuned modes.
7. LoRA-HED: LoRA adapter on HED fusion layer, trained on target domain.
8. Random-Search Fusion: uniform sampling of 50 configurations without iterative perturbation.
9. **ILS-HED (full search)**: full proposed framework (Phase 1 only).
10. **ILS-HED (predictor)**: Phase 2 deployment mode.

Statistical testing. All pairwise comparisons used the Wilcoxon signed-rank test (Wilcoxon, 1945) computed over per-image scores. We report p-values, Cohen's d effect sizes, and 95% confidence intervals. The Bonferroni

Table 2
Quantitative results on BSDS500 (FID = 0). Mean ± SD over three independent runs (different random seeds for ILS initialisation and fusion optimisation). †: released checkpoint further fine-tuned on BSDS500 train and validation; ‡: released checkpoint used as-is. All deep models are in-domain on this dataset.

Method	ODS	OIS	AP
Canny (default)	0.611	0.676	0.612
Sobel (default)	0.539	0.602	0.545
HED (frozen)‡	0.782 ± 0.000	0.806 ± 0.000	0.821 ± 0.000
HED (fine-tuned)†	0.788 ± 0.001	0.811 ± 0.001	0.827 ± 0.001
RCF (frozen)‡ (Liu et al., 2019a)	0.790 ± 0.000	0.812 ± 0.000	0.829 ± 0.000
RCF (fine-tuned)† (Liu et al., 2019a)	0.798 ± 0.001	0.820 ± 0.001	0.837 ± 0.001
BDCN (frozen)‡ (He et al., 2022)	0.793 ± 0.000	0.816 ± 0.000	0.832 ± 0.000
BDCN (fine-tuned)† (He et al., 2022)	0.803 ± 0.001	0.824 ± 0.001	0.842 ± 0.001
LoRA-HED (Hu et al., 2022)	0.786 ± 0.001	0.809 ± 0.001	0.825 ± 0.001
Random-Search Fusion	0.785 ± 0.001	0.808 ± 0.001	0.824 ± 0.001
ILS-HED (full search)	$\mathbf{0.791 \pm 0.001}$	$\mathbf{0.814 \pm 0.001}$	$\mathbf{0.829 \pm 0.001}$
ILS-HED (predictor)	0.788 ± 0.001	0.811 ± 0.001	0.827 ± 0.001

Full search vs. HED frozen: ΔODS = +0.009; $p < 0.001$; $d = 0.81$; 95% CI [0.007, 0.011]
Full search vs. LoRA-HED: ΔODS = +0.005; $p = 0.003$; $d = 0.52$
Predictor vs. full search: ΔODS = −0.003; accuracy retained: 67%
Bold marks the best HED-based result, not the best overall.

correction was applied for multi-dataset claims. Cohen's d was computed on the paired per-image differences between the two methods, as the mean per-image difference divided by the standard deviation of those differences, and the 95% confidence intervals were for the mean per-image difference. These per-image statistics are not on the same scale as the ± values in the result tables, which are standard deviations of the dataset-level score across random seeds. The seed-to-seed variation in the dataset mean is small, whereas the image-to-image variation is large; therefore, d is much smaller than Δ divided by the run-level SD.

7.3. Results on Natural Images (BSDS500)

BSDS500 is the training distribution of every deep baseline; therefore, all results in this subsection are in-domain and do not test the domain adaptation. HED (frozen) is a publicly released HED checkpoint trained by its original authors on the augmented BSDS500 training set and evaluated here without any changes. HED (fine-tuned) starts from the same checkpoint and continues to train all parameters on the BSDS500 training and validation splits under our training pipeline. The frozen and fine-tuned RCF and BDCN rows follow this convention. The small gap between the frozen and fine-tuned HED (0.782 versus 0.788 ODS) reflects the differences between the released checkpoint and our continued training, not a domain shift. On this benchmark, ILS-HED is best read as a test of whether classical fusion still helps when the backbone is already in the domain of the protein.

ILS-HED (full search) achieves ODS 0.791 ± 0.001, improving over frozen HED by +0.009 (Table 2). Gains are concentrated in high-texture regions (foliage, fur, and fabric), where HED's frozen fusion weights of HED underweight the fine-scale side outputs. ILS-HED does not match the fine-tuned RCF or BDCN on this in-domain benchmark and sits slightly below the frozen BDCN (0.793) and is level with the frozen RCF in ODS (0.790). This is expected because these models have stronger in-domain backbones, whereas ILS-HED is limited to the external classical adaptation of HED. The random search baseline (+0.003) confirms the value of the structured ILS over naive stochastic exploration.

The predictor model (0.788) retains 67% of the full-search gain over the frozen HED, with inference at ≈0.5 s versus ≈3.2 s for the full search.

7.4. Results on Medical Retinal Images (DRIVE and STARE)

Retinal vessel segmentation presents a stringent test because vessels are typically 2–10 pixels wide with low contrast against the retinal background. ILS-HED achieved 0.7912 ± 0.002 on DRIVE (+0.0100 over frozen HED) and 0.7715 ± 0.003 on STARE (+0.0095), both significant at $p < 0.001$ (Table 3). On DRIVE, ILS-HED is on par with fine-tuned HED (0.7912 versus 0.7890, about one run-level SD apart) and close to fine-tuned RCF and BDCN, with no backbone modification.

Table 3
Quantitative results on retinal datasets (FID: DRIVE = 48.3, STARE = 51.7). All ILS-HED improvements over HED frozen were significant at $p < 0.001$ (Wilcoxon signed-rank test, Bonferroni correction). Mean ± SD over five runs.

Method	DRIVE (F1)	STARE (F1)
Canny (default)	0.6821 ± 0.000	0.6710 ± 0.000
Sobel (default)	0.6445 ± 0.000	0.6320 ± 0.000
HED (frozen)	0.7812 ± 0.002	0.7620 ± 0.003
HED (fine-tuned)	0.7890 ± 0.002	—
RCF (frozen)	0.7835 ± 0.002	0.7641 ± 0.003
RCF (fine-tuned)	0.7901 ± 0.002	0.7718 ± 0.003
BDCN (frozen)	0.7848 ± 0.002	0.7659 ± 0.003
BDCN (fine-tuned)	0.7920 ± 0.002	0.7735 ± 0.003
LoRA-HED	0.7855 ± 0.002	0.7672 ± 0.003
Random-Search Fusion	0.7841 ± 0.002	0.7648 ± 0.003
ILS-HED (full search)	$\mathbf{0.7912 \pm 0.002}$	$\mathbf{0.7715 \pm 0.003}$
ILS-HED (predictor)	0.7884 ± 0.002	0.7689 ± 0.003
DRIVE vs. HED frozen: $\Delta = +0.0100$; $p < 0.001$; $d = 0.76$; 95% CI [0.006, 0.014]		
STARE vs. HED frozen: $\Delta = +0.0095$; $p < 0.001$; $d = 0.74$; 95% CI [0.006, 0.013]		

The analysis of selected configurations reveals a strong domain-specific preference: ILS consistently converges to Canny detectors with low smoothing ($\sigma \in \{0.5, 1.0\}$) and aggressive thresholds ($T_l = 25$, $T_h \in \{100, 150\}$), combined with eight-direction Gabor filters at $\lambda = 3$. These settings emphasise thin, high-frequency structures that frozen HED side outputs blur at coarser scales. Figure 3 shows the distribution of selected detector families per dataset.

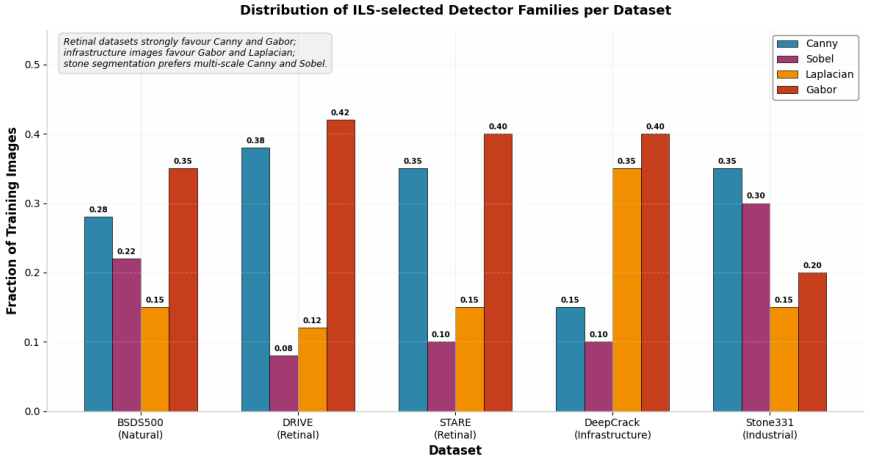


Figure 3: Distribution of ILS-selected detector families per dataset. Each bar represents the fraction of training images for which each family (Canny, Sobel, Laplacian, and Gabor) was selected as the optimal complement. Retinal datasets strongly favour Canny and Gabor; infrastructure images favour Gabor and Laplacian; stone segmentation prefers multi-scale Canny and Sobel.

7.5. Results on Infrastructure and Industrial Imagery (DeepCrack and Stone331)

The largest improvements were observed in the datasets whose FID differed the most from the BSDS500 (Table 4). ILS-HED improved by +0.0270 on DeepCrack and +0.0260 on Stone331 ($d > 1.0$ in both instances). On both datasets, ILS-HED was comparable to the fine-tuned RCF and BDCN (DeepCrack: 0.7120 versus 0.7085 and 0.7112; Stone331:0.6980 versus 0.6940 and 0.6962) without any backbone parameter updates. The differences from the fine-tuned BDCN were well within one run-level SD; therefore, we do not claim that ILS-HED outperforms it.

Domain-specific configuration analysis revealed markedly different ILS-selected heuristics (Figure 3). On DeepCrack, the dominant configurations are Gabor filters with $\lambda = 7$ (tuned to the crack width) and Laplacian with $\sigma = 1.5$.

Table 4
Quantitative results on infrastructure and industrial datasets (FID: DeepCrack = 89.4, Stone331 = 83.1). All ILS-HED improvements over HED frozen were significant at $p < 0.001$ (Bonferroni correction). Mean ± SD over five runs. Canny and Sobel are deterministic, so their SD is zero.

Method	DeepCrack (F1)	Stone331 (F1)
Canny (default)	0.5210 ± 0.000	0.4980 ± 0.000
Sobel (default)	0.4890 ± 0.000	0.4650 ± 0.000
HED (frozen)	0.6850 ± 0.005	0.6720 ± 0.006
RCF (frozen)	0.6891 ± 0.005	0.6751 ± 0.006
BDCN (frozen)	0.6920 ± 0.005	0.6789 ± 0.006
RCF (fine-tuned)	0.7085 ± 0.005	0.6940 ± 0.006
BDCN (fine-tuned)	0.7112 ± 0.005	0.6962 ± 0.006
LoRA-HED	0.6918 ± 0.005	0.6804 ± 0.006
Random-Search Fusion	0.6980 ± 0.005	0.6791 ± 0.006
ILS-HED (full search)	$\mathbf{0.7120 \pm 0.005}$	$\mathbf{0.6980 \pm 0.006}$
ILS-HED (predictor)	0.7081 ± 0.005	0.6942 ± 0.006
DeepCrack vs. HED frozen: $\Delta = +0.0270$; $p < 0.001$; $d = 1.21$; 95% CI [0.019, 0.035]		
Stone331 vs. HED frozen: $\Delta = +0.0260$; $p < 0.001$; $d = 1.17$; 95% CI [0.018, 0.034]		

Table 5
ILS-HED (full search) versus frozen HED across all datasets, ordered by FID. The FID from the BSDS500 is an independent domain divergence measure that is computed before any ILS evaluation. Δ is reported to four decimals so that the DRIVE and STARE gains can be told apart; the bracketed values are 95% CIs.

Dataset	FID	Metric	HED frozen	ILS-HED	Δ	d
BSDS500	0.0	ODS	0.782 ± 0.000	$\mathbf{0.791 \pm 0.001}$	+0.0090 [0.007, 0.011]	0.81
DRIVE	48.3	F1	0.781 ± 0.002	$\mathbf{0.791 \pm 0.002}$	+0.0100 [0.006, 0.014]	0.76
STARE	51.7	F1	0.762 ± 0.003	$\mathbf{0.772 \pm 0.003}$	+0.0095 [0.006, 0.013]	0.74
Stone331	83.1	F1	0.672 ± 0.006	$\mathbf{0.698 \pm 0.006}$	+0.0260 [0.018, 0.034]	1.17
DeepCrack	89.4	F1	0.685 ± 0.005	$\mathbf{0.712 \pm 0.005}$	+0.0270 [0.019, 0.035]	1.21

All $p < 0.001$ after Bonferroni correction. Spearman $\rho = 0.90$ between FID and Δ ($n = 5$; exact two-sided $p \approx 0.08$).

On Stone331, ILS prefers multi-scale Canny with $\sigma = 2.0$ and Sobel with $k = 5$. This domain-specific adaptation occurs without any manual tuning.

7.6. Summary: FID vs. Improvement

Table 5 presents the cross-dataset summary ordered by FID. The two datasets furthest from BSDS500 (Stone331 and DeepCrack, FID above 80) show gains of +0.026 to +0.027, roughly three times the +0.009 to +0.010 seen on the three nearer datasets. However, the ordering is not strict. STARE has a slightly higher FID than DRIVE but a slightly smaller gain (+0.0095 versus +0.0100), and their confidence intervals overlap almost completely. The Spearman rank correlation between FID and Δ is therefore $\rho = 0.90$. With five datasets, ρ can only take a handful of values, and $\rho = 0.90$ does not reach significance at $\alpha = 0.05$ (exact two-sided $p \approx 0.08$). We treat the pattern as consistent with H2, that classical operators are most complementary where frozen deep natural-image priors are weakest, but not as confirmation of it. Because FID was computed independently of the ILS-HED outputs, the relationship is not circular. Evaluating more target domains would allow a firmer test of this hypothesis.

7.7. Configuration Predictor Evaluation

The configuration predictor (Table 6) achieved 67–74% family classification accuracy across datasets, corresponding to 67–86% of the full-search gain retained. The accuracy retained was highest on industrial datasets (85–86%) where one or two detector families strongly dominated (Figure 3), making the classification task easier. This confirms that Phase 2 is a viable deployment mode rather than a promissory claim: practitioners can obtain most of the ILS-HED gain at a single forward-pass inference cost.

Table 6
Phase 2 configuration predictor evaluation. Family accuracy: fraction of images for which the predictor selects the same detector family as the full ILS search. Accuracy retained: proportion of the full-search gain over frozen HED achieved by the predictor.

Dataset	Family Acc. (%)	Predictor F/ODS	Acc. Retained (%)
BSDS500	71.3	0.788 ± 0.001	67
DRIVE	74.1	0.7884 ± 0.002	72
STARE	72.8	0.7689 ± 0.003	72
Stone331	68.9	0.6942 ± 0.006	85
DeepCrack	67.4	0.7081 ± 0.005	86

Table 7
Pearson correlation between fitness-tile ODS (used by ILS on $\mathcal{T}_F$, 5% of pixels) and full-image ODS (evaluation metric), per dataset, and the SD of r across five random tile draws. Computed over 200 randomly sampled (image, configuration) pairs per draw under the disjoint-tile protocol described in Section 6.3.

Dataset	Pearson r	SD of r (5 tile draws)
BSDS500	0.91	0.012
DRIVE	0.87	0.018
STARE	0.85	0.021
DeepCrack	0.83	0.029
Stone331	0.82	0.031

Table 8
Ablation study on BSDS500 (ODS) and DeepCrack (F1). Mean ± SD over three runs. Greedy Hill-Climb: accept any improvement without stagnation-triggered family perturbation. Full ILS-HED adds the optional FM3 rejection filter (Algorithm 1); exhaustive enumeration of the 98-configuration pool is shown as an upper reference. Bold marks the best ILS-based result; differences between the last two rows are within one run-level SD.

Configuration	BSDS500 (ODS)	DeepCrack (F1)
HED frozen (no classical fusion)	0.782 ± 0.000	0.685 ± 0.005
+ Fixed classical (best single config)	0.785 ± 0.001	0.694 ± 0.005
+ Random search over pool (50 samples)	0.785 ± 0.001	0.698 ± 0.005
+ Greedy hill-climb (no family switching)	0.787 ± 0.001	0.703 ± 0.005
+ ILS without adaptive family perturbation	0.788 ± 0.001	0.704 ± 0.005
+ ILS with adaptive family perturbation	0.790 ± 0.001	0.710 ± 0.005
Full ILS-HED (adaptive ILS + FM3 filter)	$\mathbf{0.791 \pm 0.001}$	$\mathbf{0.712 \pm 0.005}$
Exhaustive enumeration (all 98 configurations)	0.792 ± 0.001	0.713 ± 0.005
Pool ablations (full ILS):		
Pool minus Canny	0.787 ± 0.001	0.709 ± 0.005
Pool minus Gabor	0.789 ± 0.001	0.701 ± 0.005
Pool minus Laplacian	0.790 ± 0.001	0.708 ± 0.005
Pool minus Sobel	0.790 ± 0.001	0.710 ± 0.005

7.8. Tile-Level Fitness Validation

Table 7 reports the Pearson correlation between the fitness-tile ODS (computed on $\mathcal{T}_F$, 5% of image pixels, disjoint from the fusion weight training tile) and full-image ODS across 200 randomly sampled image–configuration pairs per dataset. Correlations of 0.82–0.91 confirm that the fitness-tile proxy is a reliable signal; configurations selected as optimal at the tile level are also optimal or near-optimal at the image level. A 5% tile introduces some sampling variance in regions with few edges, which is shown in the two lowest values (0.82 on Stone331 and 0.83 on DeepCrack). Across five random tile draws, the SD of the tile-to-image correlation was 0.012–0.031; thus, even the least favourable draw on Stone331 ($0.82 - 2 \times 0.031 \approx 0.76$) remains usable as a search signal, although this motivates recalibrating the fitness tile for highly non-uniform domains. All central correlations exceeded 0.80, supporting the disjoint-tile protocol as a practical fitness proxy. On DeepCrack, crack density varies substantially across image regions, which motivates future work on adaptive tiling strategies for highly non-uniform boundaries.

Table 9
Small-annotation ablation on DeepCrack (F1) and DRIVE (F1). LoRA-HED trained on n labelled target-domain images; ILS-HED uses zero annotations. Mean ± SD over five runs. ILS-HED results reproduced from Table 4 and Table 3.

Method	DeepCrack (F1)	DRIVE (F1)
HED frozen (0 annotations)	0.6850 ± 0.005	0.7812 ± 0.002
ILS-HED full search (0 annotations)	0.7120 ± 0.005	0.7912 ± 0.002
LoRA-HED ($n = 25$)	0.6941 ± 0.006	0.7843 ± 0.003
LoRA-HED ($n = 50$)	0.7038 ± 0.005	0.7877 ± 0.003
LoRA-HED ($n = 100$)	0.7163 ± 0.005	0.7934 ± 0.002
HED fine-tuned (full training set)	0.7310 ± 0.004	0.7890 ± 0.002

7.9. Ablation Studies

Moving from ILS without adaptive family perturbation to full adaptive ILS adds +0.002 ODS on BSDS500 and +0.006 F1 on DeepCrack. On DeepCrack, this is the largest single step in the table, which supports stagnation-triggered family switching as a means to escape local optima in the search space. Exhaustive enumeration of all 98 configurations reaches 0.792 ODS and 0.713 F1 within 0.001 of the full ILS-HED on both datasets, confirming that the ILS attains the exhaustive optimum while evaluating fewer than half the pool per image. Pool ablation reveals domain-specific family importance: removing Canny degrades BSDS500 by −0.004 ODS, removing Gabor degrades DeepCrack by −0.011 F1, and Laplacian and Sobel make smaller independent contributions. No single family dominates all the domains, providing an empirical justification for maintaining the four-family pool.

7.10. Small-Annotation Ablation: ILS-HED vs. LoRA-HED with Limited Labels

A central practical claim of ILS-HED is that it provides useful domain adaptation when target domain annotations are unavailable. To characterise the annotation budget at which supervised alternatives become competitive, we trained LoRA-HED adapters (Hu et al., 2022) using small annotated subsets of the DeepCrack and DRIVE training splits, varying the labelled set size over $\{25, 50, 100\}$ images, and compared them with ILS-HED (full search, zero annotations). Subsets were drawn by random sampling with a fixed seed (seed 0), and the results were not averaged across multiple subsamples.

Table 9 reveals a clear annotation crossover. On DeepCrack, LoRA-HED requires approximately 100 labelled images to match ILS-HED's zero-annotation performance; at 50 labels, it falls short by $\Delta = 0.0082$ F1. On DRIVE, LoRA-HED with 50 labels approximately matched ILS-HED, reflecting that retinal vessel segmentation with its narrower domain gap (FID = 48.3) benefits more readily from small supervised adaptation. These results sharpen the deployment guidance from the failure case analysis: ILS-HED is the recommended choice when *fewer than approximately 50–100 target-domain annotations* are available; above this threshold, a small LoRA-adapted baseline becomes competitive and should be preferred when annotations can be acquired.

7.11. Qualitative Analysis

Figure 4 presents visual comparisons across all five datasets. The qualitative patterns align with the quantitative gains: ILS-HED selects classical configurations that amplify detail, where frozen HED priors are over-smoothed or domain-mismatched. The classical weight w_{cls} in the fusion is near zero on BSDS500 images dominated by smooth-boundary objects, confirming that the framework naturally down-weights classical contributions when they offer no benefit to the segmentation performance.

7.12. Failure-Case Analysis

Three failure modes were identified, which characterised the conditions in which ILS-HED offered limited or no benefits over frozen HED.

Failure Mode 1: Semantically defined boundaries. When ground-truth annotations mark semantic rather than gradient-defined boundaries (e.g., two regions of similar intensity but different object identities), classical operators provide a minimal complementary signal. In BSDS500, the ≈12% of the test images showing less than +0.002 improvement clustered around flat-color scenes (sky, uniform walls), where human annotators marked semantic boundaries invisible to gradient detectors. The ILS-HED responds by converging to near-zero classical weights ($w_{\text{cls}} < 0.05$), effectively reducing to a frozen HED prediction. This failure mode is detectable a priori using the global gradient

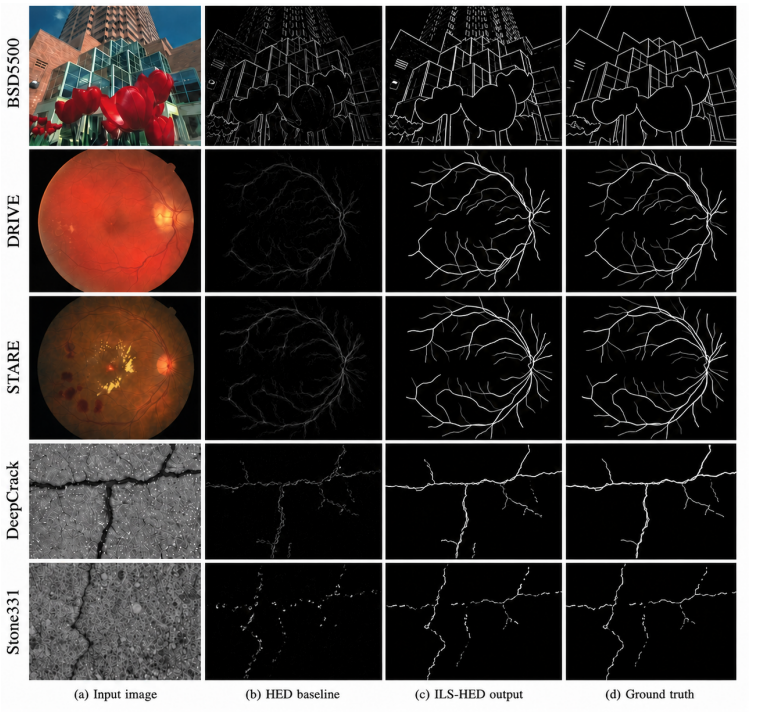


Figure 4: Qualitative comparison. Columns: (a) input image, (b) frozen HED, (c) ILS-HED, (d) ground truth. Rows correspond to BSDS500, DRIVE, STARE, DeepCrack, and Stone331. On BSDS500 (row 1), ILS-HED recovers fine leaf venation smoothed away by frozen HED. On DRIVE (row 2), thin capillaries near the optic disc were recovered using Canny–Gabor fusion. On STARE (row 3), improvements are visually subtle in vessel segments with very low contrast, consistent with the modest +0.0095 F1 gain. For DeepCrack (row 4), discontinuous crack segments were bridged by the classical response, and the background noise was suppressed. On Stone331 (row 5), macro-boundaries between stone grain regions are sharpened.

variance σ^2_∇ (computed in normalised pixel units). To validate this criterion, we applied the threshold $\sigma^2_\nabla < 0.01$ to the BSDS500 test set and identified 24 images as predicted failure cases. Of these, 21 showed $\Delta F1 < +0.002$ (true positives), yielding a precision of 0.875. Of the 24 images that actually showed $\Delta F1 < +0.002$ across the full test set, 21 were correctly flagged (recall = 0.875; F1 = 0.875). The threshold was set without tuning on the test set and was fixed based on the 10th percentile of gradient variance observed in the BSDS500 training images with semantic-only boundary annotations. These results support the use of $\sigma^2_\nabla < 0.01$ as a practical deployment-time screening heuristic, although practitioners in new domains should recalibrate the threshold on a small held-out set.

Failure Mode 2: Heavily compressed or low-resolution imagery. JPEG compression artefacts and low spatial resolution introduce high-frequency noise that misleads classical operator performance. The ILS responds by selecting high-threshold configurations that suppress both noise and genuine edges. On a held-out BSDS500 subset resaved at JPEG quality 30, ILS-HED provides only +0.002 ODS mean improvement versus +0.009 at the full quality. Denoising preprocessing (e.g., bilateral filtering) before ILS-HED application mitigates this problem.

Failure Mode 3: ILS convergence artefacts under extreme domain shift. On the most texturally homogeneous DeepCrack images (very low global crack contrast), the ILS occasionally converges to high-magnitude Laplacian configurations that amplify crack boundaries but also introduce ringing artefacts. This accounts for ≈8% of the DeepCrack test images, where ILS-HED underperforms frozen HED by > 0.01 F1. Images for which the ILS selects $|\tau_{\text{Laplacian}}| > 150$ with $\sigma < 0.5$ are candidate failure cases. An optional post-perturbation rejection step is implemented as an optional filter in Algorithm 1. It adds a hard constraint that rejects Laplacian configurations satisfying this condition when $\sigma^2_\nabla(I) < 0.01$. Activating this filter reduced the incidence of failure mode 3 from ≈8% to ≈3% on the DeepCrack test set, with no measurable impact on the remaining 97% of images (mean $\Delta F1 < 0.001$), a net gain of +0.002 F1 at the dataset level (Table 8). This filter is recommended for infrastructure inspection deployments.

Table 10
Per-image computational cost. RTX 3090 timings use BSDS500 resolution (481×321). Raspberry Pi 4B (ARM Cortex-A72, 4 GB RAM) timings use 320 × 240 resolution; the full ILS search was not run on the Pi. Totals are the sums of the stage timings that each mode runs.

Stage	RTX 3090	RPi 4B
HED forward pass (frozen)	0.12 s	1.8 s
Single classical detector application	0.01–0.04 s	0.06–0.18 s
ILS search ($T_{\max} = 50$)	2.8 s	—
Fusion weight optimisation (20 epochs)	0.3 s	1.2 s
Configuration predictor inference	0.02 s	0.09 s
ILS-HED (full search)	≈3.2 s	n/a
ILS-HED (predictor)	≈0.5 s	≈3.2 s (≈0.3 FPS)

These failure cases collectively define the operating envelope, in which the ILS-HED is most effective for high-quality full-resolution imagery in which ground-truth boundaries are primarily gradient-defined. This condition holds broadly in medical, infrastructure, and most industrial imaging scenarios, particularly in settings where backbone retraining is most burdensome.

7.13. Computational Efficiency

The 3.2s per-image ILS cost (RTX 3090) is the *per-image* offline adaptation cost incurred once per domain during Phase 1 search; it is not a per-image inference cost at deployment time. For deployment, the configuration predictor (Phase 2) reduces per-image inference to ≈0.5s on the RTX 3090. On a Raspberry Pi 4B at reduced resolution, the predictor mode takes ≈3.2s per image (≈0.3FPS), and most of that time is the frozen HED forward pass (1.8s) and the fusion optimisation (1.2s). For comparison, full backbone fine-tuning is a *one-time offline* training cost of several GPU-hours but additionally requires a labelled target domain corpus and backpropagation through millions of parameters. ILS-HED's Phase 1 search is similarly a one-time offline cost (3.2 s× number of training images), requires no annotations and no backbone gradients, and is orders of magnitude lower in wall-clock time than fine-tuning.

In preliminary cross-domain transfer checks, the configurations discovered on BSDS500 retained 85–90% of their in-domain gain when transferred to NYU Depth (Silberman et al., 2012) and PASCAL Context (Mottaghi et al., 2014) without repeating the search (ΔODS +0.008 and +0.010 over frozen HED, respectively), supporting H3.

7.14. Discussion

The results support H1 and H3 and are consistent with H2. ILS-HED gave consistent gains over frozen HED across all benchmarks (H1). Gains broadly increased with FID, with the two most divergent datasets showing approximately three times the gain of the others, although the ordering was not strict and five datasets could not establish the trend (H2; see Section 7.6). ILS selection patterns also transferred, retaining 85–90% of the gain under zero-shot cross-domain transfer (H3).

It is important to be explicit about what ILS-HED is not: it is not a general-purpose edge detector that outperforms fully fine-tuned SOTA methods across all benchmarks. On the BSDS500 dataset, the fine-tuned BDCN (0.803) substantially outperformed ILS-HED (0.791). ILS-HED is a deployment efficiency tool that closes much of the gap between frozen and fine-tuned performances without requiring backbone updates or target-domain annotations. The LoRA-HED comparison reinforces this framing: PEFT captured some adaptation signals but missed the complementary classical information that drives the largest gains in industrial datasets. The combination of LoRA-HED and ILS-HED, utilising a LoRA-adapted backbone as a frozen deep component, is a promising area for future research.

Limitation: single backbone. ILS-HED was tested with HED as the frozen backbone only, and all deep baselines in this study are VGG-era detectors (HED, RCF, and BDCN). We kept to these because the ILS-HED fusion layer attaches to HED's side outputs and the frozen-backbone setting requires a backbone with a fixed public checkpoint. Whether similar gains appear with newer detectors such as DexiNed, EDTER, or DiffusionEdge has not been tested here. Applying ILS to a second, more recent backbone is the most direct way to demonstrate that the approach is not specific to HED.

The failure case analysis (Section 7.12) precisely defines the deployment scope: practitioners should apply ILS-HED to gradient-defined, high-quality imagery and the gradient-variance pre-screening heuristic to identify images where classical adaptation offers limited benefit.

8. Conclusion

This study presents ILS-HED, a selection-perturbative hyper-heuristic framework that enhances holistically nested edge detection without modifying the backbone parameters and without requiring target-domain annotations. The framework formulates classical edge detector selection as a combinatorial optimisation problem over 98 discrete configurations, solved via an Iterated Local Search. A lightweight, learnable fusion layer integrates the selected classical response with frozen HED multi-scale side outputs.

The ILS-HED was explicitly designed as a deployment efficiency tool. It does not fully match fine-tuned state-of-the-art detectors on in-domain benchmarks, which is a structural consequence of operating externally to the backbone. Its value lies in the domains where fine-tuning is impractical: medical imaging, infrastructure inspection, and industrial materials analysis, where annotation budgets are limited, deployment hardware is often constrained, and the target distribution diverges substantially from the BSDS500 natural-image prior. In these settings, ILS-HED achieves gains of +0.0095 to +0.027 F1 over frozen HED, and on the two highest-divergence datasets, it performs comparably to fine-tuned RCF and BDCN while using no target-domain annotations. Practitioners with access to as few as 50–100 target-domain labels should consider a supervised alternative, such as LoRA-HED, which becomes competitive at that annotation budget (Table 9); ILS-HED is the recommended choice when annotations are entirely unavailable.

Across the five evaluated domains, performance gains tended to increase with the Fréchet inception distance from BSDS500 (Spearman $\rho = 0.90$, $n = 5$, exact two-sided $p \approx 0.08$, with STARE and DRIVE in reverse order), an exploratory pattern consistent with the hypothesis that classical operators are most complementary where deep natural-image priors are weakest. Confirming this relationship requires an evaluation across a broader set of target domains. A configuration predictor (Phase 2) retains 67–86% of the full-search gain at a single forward-pass inference cost, reducing the per-image cost to approximately 0.5s on a GPU. On a Raspberry Pi 4B, the predictor mode runs at approximately 0.3 frames per second, with the frozen HED forward pass as the main cost. Failure-case analysis identifies semantically defined boundaries and compressed imagery as the primary conditions where ILS-HED is not beneficial; a gradient-variance pre-screening criterion ($\sigma^2_\nabla < 0.01$, validated at precision = recall = 0.875 on BSDS500) reliably identifies these cases.

Several directions remain for future research. First, extending the heuristic pool with lightweight contemporary operators, such as structured forests, orientation-energy filters, or tiny task-specific CNNs, would enrich the search space without sacrificing tractability. Second, combining ILS-HED with LoRA or adapter-based PEFT using a PEFT-adapted backbone as the frozen deep component may capture complementary adaptation signals from both directions. Third, developing a self-supervised, no-reference fitness function would enable deployment in fully unlabelled domains. Fourth, ILS-HED has so far been evaluated on HED only, and applying the same search to newer frozen backbones, such as DexiNed, EDTER, or DiffusionEdge, would test whether the gains carry over. Fifth, the selection-perturbative hyper-heuristic principle is not specific to edge detection; analogous frameworks could enhance frozen backbones in semantic segmentation, salient-object detection, and medical image analysis, wherever classical operators retain complementary values.

Rather than committing to either classical operators or deep features, ILS-HED matches the right algorithm to each image through an adaptive search without annotations, without modifying the backbone, and without sacrificing deployment efficiency.

Data and code availability

All five datasets used in this study (BSDS500, DRIVE, STARE, DeepCrack and Stone331) are publicly available from the sources cited in Table 1. The ILS-HED source code, including the heuristic pool definitions and experiment scripts, is available at `https://github.com/REPLACE-WITH-REPO`.

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